

Elementary bounds on digital sums of powers

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Some notation

For integers $N \geq 0$ and $b \geq 2$, the *base- b expansion* of N is the unique representation of the form

$$N = \sum_{i=0}^{\infty} d_i b^i, \quad d_i \in \{0, 1, \dots, b-1\}.$$

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We write $s_b(N)$ for the sum of the base- b digits of N , and $c_b(N)$ for the number of nonzero digits in that expansion.

A question to start with

Does the sum of decimal digits of 2^n increase to infinity with n ?

n	2^n	$s_{10}(2^n)$
5	32	5
10	1024	7
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It looks like $\lim_{n \rightarrow \infty} s_{10}(2^n) = \infty$, but the sequence is not monotonic.

The last digit of 2^n

We write the decimal expansion of 2^n as

$$2^n = \sum_{i=0}^{\infty} d_i 10^i,$$

where each digit satisfies $d_i \in \{0, 1, \dots, 9\}$.

First easy fact

The last digit d_0 is nonzero, because a power of 2 is never divisible by 10.

So every 2^n already has *at least one* nonzero digit.

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But the only multiple of 16 below 10 is 0, and our remainder is not 0 because $d_0 \neq 0$.

Conclusion

At least one of the digits d_1, d_2, d_3 must be nonzero.

A bigger block gives a new nonzero digit

Now assume $n \geq 14$. Then 2^n is divisible by 2^{14} , and

$$2^{14} = 16384 > 10^4.$$

Look at the last 14 digits:

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This remainder is divisible by 2^{14} .

Now suppose the ten digits $d_4, \dots, d_{13}$ were all zero. Then the last 14 digits would look like

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so the remainder would be less than 10^4 , which is a contradiction.

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Conclusion

At least one digit in the block $d_4, d_5, \dots, d_{13}$ is nonzero.

What is the pattern?

We keep choosing larger and larger cutoffs:

$$4, 14, 47, 157, \dots$$

so that each new power of 2 is bigger than the previous power of 10.
For example,

$$2^{14} > 10^4, \quad 2^{47} > 10^{14}, \quad 2^{157} > 10^{47}.$$

Then the same argument says:

- $n \geq 4 \implies$ block d_1, d_2, d_3 has a nonzero digit,
- $n \geq 14 \implies$ block $d_4, \dots, d_{13}$ has a nonzero digit,
- $n \geq 47 \implies$ block $d_{14}, \dots, d_{46}$ has a nonzero digit,
- $n \geq 157 \implies$ block $d_{47}, \dots, d_{156}$ has a nonzero digit, etc.

Why this works

As n grows, more and more of these disjoint blocks become available.

Each block contains *at least one* nonzero digit.

So the number of nonzero digits of 2^n tends to infinity.

Consequently, the sum of digits of 2^n also tends to infinity.

Generalization to other bases

Since we are using no special properties of 2 and 10 (except $2 \mid 10$) we can generalize this result beyond $a = 2$ and $b = 10$.

Theorem (Radcliffe, 2025)

Let $2 \leq a < b$ with $a \mid b$. Suppose N is divisible by a^n but not b . Then there exists $C > 0$ (depending on a, b) such that

$$c_b(N) > C \log n$$

for n sufficiently large.

Handling General Bases

The condition that $a \mid b$ can be weakened considerably.

Corollary

Let $a, b \geq 2$. Let d be the smallest factor of a with $\gcd(a/d, b) = 1$. If $\log(d)/\log(b)$ is irrational, then

$$c_b(a^n) > C \log n$$

for sufficiently large n .

This irrationality condition ensures that after removing powers of b from a^n , the remainder has a prime factor with large p -adic valuation.

Example: $a = 6$ and $b = 12$.

Stewart's Theorem

Theorem (Stewart, 1980)

Let $a, b \geq 2$ with $\log(a)/\log(b)$ irrational. Then

$$s_b(a^n) \geq c_b(a^n) > \frac{\log n}{\log \log n + C} - 1$$

for all $n > 4$, where C depends only on a and b .

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Note: The irrationality condition is necessary!

If $\log(a)/\log(b) = r/s \in \mathbb{Q}$, then

$$a^{ns} = b^{nr}$$

has only one nonzero digit in base b .

- The condition of Stewart's theorem is much weaker than ours.
- However, the conclusion is also weaker.
- Our results require only elementary number theory.
- Stewart's theorem requires advanced methods.

Baker's Theorem

The proof of Stewart's theorem requires a deep result from transcendental number theory:

Theorem (Baker-Wüstholz, 1993)

Let $\Lambda = b_1 \log \alpha_1 + \cdots + b_n \log \alpha_n$ where:

- $b_1, \dots, b_n$ are integers with $|b_i| \leq B$
- $\alpha_1, \dots, \alpha_n$ are algebraic with heights $\leq A_i$

If $\Lambda \neq 0$, then

$$\log |\Lambda| > -(16nd)^{2(n+2)} \log A_1 \cdots \log A_n \log B$$

where $d = [\mathbb{Q}(\alpha_1, \dots, \alpha_n) : \mathbb{Q}]$.

Proof Strategy

Let

$$a^n = b^m (d_1 b^{-m_1} + \dots + d_k b^{-m_k}),$$

where $m = \lceil \log_b a^n \rceil$, $k = c_b(a^n)$, $d_i \in \{1, \dots, b-1\}$, and

$$1 = m_1 < \dots < m_k \leq m.$$

Write a^n in base b and truncate at the i -th nonzero digit:

$$a^n = b^{m-m_i} q_i + r_i$$

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Goal: Show digit gaps m_{i+1}/m_i cannot be too large

Main steps of proof

- 1 Approximate a^n by truncating to the i -th nonzero digit.

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- 4 Deduce thereby an upper bound on the digit gap.

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- 3 Apply Baker's theorem to obtain a lower bound on the truncation error.
- 4 Deduce thereby an upper bound on the digit gap.
- 5 Compute a lower bound on the number of gaps, and hence the number of nonzero digits.